

ChatGPT

Home

Day8

Day10

Download file | iLovePDF

github login - Search

gayuaid016-ops/DataSci

localhost:8888/notebooks/Workspace/Day8.ipynb?

28°C Cloudy

Day8 Last Checkpoint: 3 days ago

File Edit View Run Kernel Settings Help

JupyterLab Python [conda env:base] * Trusted

[1]:

```
#Outliers
#An outliers is a data point that issignificantly differentt from the rest of the data
#Outliers can: (Disort the mean,affect standard deviation,reduce model performance * influence scaling and regression models)
#Method 1:Visual Detection Using Boxplot
#it quickly shows
#1.Minimum value,2.Q1,3.median,4.Q3,maximum value,outliers
#Method 2:IQR Method
#Step-1:Calculate IQR,Step2:Calculate Limits-Lower limit/&-upperLimit,Step3:Detect Outliers
#Method 3:Z-Score Method
#Measures how many standard deviation a values is from the mean
#Step 1:Calculate Z-Score,Step2:Detect Outliers
```

[6]:

```
import seaborn as sns
import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
from scipy.stats import zscore
df=sns.load_dataset("titanic")
#Box plot
sns.boxplot(x=df['fare'])
plt.show()
#IQR Method
Q1=df['fare'].quantile(0.25)
Q3=df['fare'].quantile(0.75)
IQR=Q3-Q1
lower=Q1-1.5*IQR
upper=Q3+1.5*IQR
outliers_iqr=df[
(df['fare']< lower) |
```

12:39 PM

6/9/2026

ChatGPT

Home

Day8

Day10

Download file | iLovePDF

github login - Search

gayuaid016-ops/DataSci

localhost:8888/notebooks/Workspace/Day8.ipynb?

Chat

Anaconda Toolbox

v4.20.0

Your Toolbox for Python Projects

Anaconda Cloud

Create a New Project

Create a New Notebook

My Projects

Code Snippets

Manage Code Snippets

Environments

Create new Environment

Anaconda AI Assistant

Jupyter Day8 Last Checkpoint: 3 days ago

File Edit View Run Kernel Settings Help

JupyterLab Python [conda envbase]

```
from scipy.stats import zscore
df=sns.load_dataset("titanic")
#Box plot
sns.boxplot(x=df['fare'])
plt.show()
#IQR Method
Q1=df['fare'].quantile(0.25)
Q3=df['fare'].quantile(0.75)
IQR=Q3-Q1
lower=Q1-1.5*IQR
upper=Q3+1.5*IQR
outliers_iqr=df[
    (df['fare']< lower) |
    (df['fare']> upper)]
print("IQR Outliers:",len(outliers_iqr))
#Z_score Method
df['fare_zscore']=zscore(df['fare'])
outliers_z=df[
    abs(df['fare_zscore'])>3
]
print("Z-score Outliers:",len(outliers_z))
```

Snipping Tool

Screenshot copied to clipboard
Automatically saved to screenshots folder.

Markup and share

28°C Cloudy

Search

12:39 PM
6/9/2026

ChatGPT

Home

Day8

Day10

Download file | iLovePDF

github login - Search

gayuaid016-ops/DataSci

localhost:8888/notebooks/Workspace/Day8.ipynb?

28°C Cloudy

12:39 PM 6/9/2026

Day8 Last Checkpoint: 3 days ago

File Edit View Run Kernel Settings Help

Code

Python [conda env:base]

0

100

200

300

400

500

fare

IQR Outliers: 116

Z-score Outliers: 20

[7]:

#EDA (Exploratory Data Analysis)

#it is the process of examining ,summarizing,and visualizing data befor building machine learning models

#why important?

1.understand the structure of the dataset

#2.identify missing values

#3.Detect outliers

#4.Understand feature distribution

#5.Find relationships between variables

#6.check ata quality issues

#7.prepare data for machine learning

#Types of EDA

#1.Univariant -Analyzing one variable at a time

#numerical feature (mean,median,standard deviation,variance,histogram,boxplot)

#categorical feature(value counts,frequency distribution,barchart)

[8]:

import pandas as pd

import seaborn as sns

import matplotlib.pyplot as plt

df=sns.load_dataset("tips")

print(df.head())

print(df.info())

print(df.columns)

Snipping Tool

Screenshot copied to clipboard

Automatically saved to screenshots folder.

Markup and share

ChatGPT

Home

Day8

Day10

Download file | ILovePDF

github login - Search

gayuaid016-ops/DataSci

localhost:8888/notebooks/Workspace/Day8.ipynb?

Anaconda Toolbox

v4.20.0

Anaconda

Your Toolbox for Python Projects

Anaconda Cloud

Create a New Project

Create a New Notebook

My Projects

Code Snippets

Manage Code Snippets

Environments

Create new Environment

Anaconda AI Assistant

Jupyter Day8 Last Checkpoint: 3 days ago

File Edit View Run Kernel Settings Help

JupyterLab Python [conda envbase]

#1.Univariate -Analyzing one variable at a time
#numerical feature (mean,median,standard deviation,variance,histogram,boxplot)
#categorical feature(value counts,frequency distribution,barchart)

[a]: import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
df=sns.load_dataset("tips")
print(df.head())
print(df.info())
print(df.columns)

	total_bill	tip	sex	smoker	day	time	size
0	16.99	1.01	Female	No	Sun	Dinner	2
1	10.34	1.66	Male	No	Sun	Dinner	3
2	21.01	3.50	Male	No	Sun	Dinner	3
3	23.68	3.31	Male	No	Sun	Dinner	2
4	24.59	3.61	Female	No	Sun	Dinner	4

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 244 entries, 0 to 243
Data columns (total 7 columns):
Column Non-Null Count Dtype

0 total_bill 244 non-null float64
1 tip 244 non-null float64
2 sex 244 non-null category
3 smoker 244 non-null category
4 day 244 non-null category
5 time 244 non-null category
6 size 244 non-null int64
dtypes: category(4), float64(2), int64(1)
memory usage: 7.4 KB
None
Index(['total_bill', 'tip', 'sex', 'smoker', 'day', 'time', 'size'], dtype='object')

28°C Cloudy

Search

12:39 PM

6/9/2026

Snipping Tool

Screenshot copied to clipboard
Automatically saved to screenshots folder.

Markup and share

ChatGPT

Home

Day8

Day10

Download file | iLovePDF

github login - Search

gayuaid016-ops/DataSci

localhost:8888/notebooks/Workspace/Day8.ipynb?

28°C Cloudy

Anaconda

Your Toolbox for Python Projects

Anaconda Cloud

Create a New Project

Create a New Notebook

My Projects

Code Snippets

Manage Code Snippets

Environments

Create new Environment

Anaconda AI Assistant

Jupyter Day8 Last Checkpoint: 3 days ago

File Edit View Run Kernel Settings Help

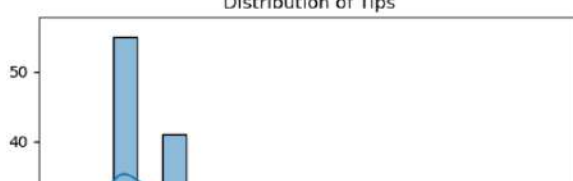
memory usage: 7.4 KB
None
Index(['total_bill', 'tip', 'sex', 'smoken', 'day', 'time', 'size'], dtype='object')

[9]:
#1.Descriptive Statistics
print(df['total_bill'].describe())

count 244.000000
mean 19.785943
std 8.902412
min 3.070000
25% 13.347500
50% 17.795000
75% 24.127500
max 50.810000
Name: total_bill, dtype: float64

[11]:
#2.Histogram
#It show the distribution of a numerical feature
#It divide data into intervals and shows how many observation fall into each interval
sns.histplot(df['tip'],bins=20,kde=True)
plt.title("Distribution of Tips")
plt.show()

Distribution of Tips



12:40 PM
6/9/2026

ChatGPT

Home

Day8

Day10

Download file | iLovePDF

github login - Search

gayuaid016-ops/DataSci

localhost:8888/notebooks/Workspace/Day8.ipynb?

28°C Cloudy

Day8 Last Checkpoint: 3 days ago

File Edit View Run Kernel Settings Help

Code

JupyterLab Python [conda env:base]

[16]:

#2. Categorical variable
print(df['day'].value_counts())

day
Sat 87
Sun 76
Thur 62
Fri 19
Name: count, dtype: int64

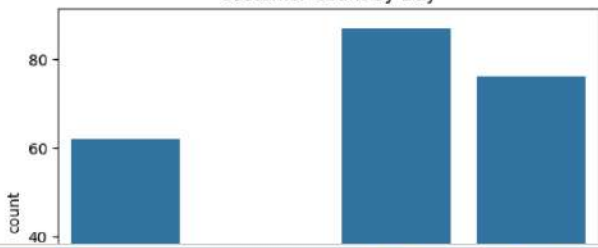
[17]:

#Count plot
sns.countplot(x='day',data=df)
plt.title("Customer Count by Day")
plt.show()

size

1 2 3 4 5 6

Customer Count by Day



Day	Count
Sat	87
Sun	76
Thur	62
Fri	19

Anaconda Toolbox

Anaconda Cloud

Code Snippets

Environments

Anaconda AI Assistant

12:40 PM

6/9/2026

ChatGPT

Home

Day8

Day10

Download file | iLovePDF

github login - Search

guyard016-ops/DataSci

localhost:8888/notebooks/Workspace/Day8.ipynb?

Anaconda Toolbox

v4.20.0

Anaconda

Your Toolbox for Python Projects

Anaconda Cloud

Create a New Project

Create a New Notebook

My Projects

Code Snippets

Manage Code Snippets

Environments

Create new Environment

Anaconda AI Assistant

Jupyter Day8 Last Checkpoint: 3 days ago

File Edit View Run Kernel Settings Help

Code

JupyterLab Python [conda env:base]

0 Paste this cell from the clipboard (V)

Thur Fri Sat Sun

day

[18]:

```
#pie chart
df['smoker'].value_counts().plot(
    kind='pie',
    autopct='%1.1f%%' #percentage value in center
)
plt.title("Smoker Percentage")
plt.ylabel("")
plt.show()
```

Smoker Percentage

No61.9%

Yes38.1%

Snipping Tool

Screenshot copied to clipboard

Automatically saved to screenshots folder.

Markup and share

28°C Cloudy

Search

12:40 PM 6/9/2026

ChatGPT

Home

Day8

Day10

Download file | iLovePDF

github login - Search

gayuaid016-ops/DataSci

localhost:8888/notebooks/Workspace/Day8.ipynb?

28°C Cloudy

Day8 Last Checkpoint: 3 days ago

File Edit View Run Kernel Settings Help

File Edit View Run Kernel Settings Help

Code

Python [conda env:base]

20

0

Thur

Fri

Sat

Sun

day

[18]:

#pie chart
df['smoker'].value_counts().plot(
kind='pie',
autopct='%1.1f%%' #percentage value in center
)
plt.title("Smoker Percentage")
plt.ylabel("")
plt.show()

Smoker Percentage

No

61.9%

12:41 PM

6/9/2026

ChatGPT

Home

Day8

Day10

Download file | iLovePDF

github login - Search

gayuaid016-ops/DataSci

localhost:8888/notebooks/Workspace/Day8.ipynb?

Anaconda

Your Toolbox for Python Projects

Anaconda Cloud

Create a New Project

Create a New Notebook

My Projects

Code Snippets

Manage Code Snippets

Environments

Create new Environment

Anaconda AI Assistant

Jupyter Day8 Last Checkpoint: 3 days ago

File Edit View Run Kernel Settings Help

JupyterLab Python [conda envbase]

Yes

```
[19]: #Outliers Detection using IQR
Q1=df['total_bill'].quantile(0.25)
Q3=df['total_bill'].quantile(0.75)
IQR=Q3-Q1
lower=Q1-1.5*IQR
upper=Q3+1.5*IQR
outliers_iqr=df[
    (df['total_bill']< lower) |
    (df['total_bill']> upper)]
print("IQR Outliers:",len(outliers_iqr))

IQR Outliers: 9

[20]: #Bivariate Analysis
#It means analyzing two variables together to understand the relationship between them
#Bi=2
#variate-variables
#Types
#1.variable1- numerical ,numerical (scatterplot)
#2.Variable2-Categorical,numerical (barchart)
#3.Visualization-Categorical,Categorical(Grouped Statistics/Crosstab)

[22]: #Numerical vs Numerical(used to determine whether one numerical variable affects another
plt.figure(figsize=(8,5))
sns.scatterplot(
    x='total_bill',
```

Snipping Tool

Screenshot copied to clipboard
Automatically saved to screenshots folder.

Markup and share

28°C Cloudy

Search

12:41 PM 6/9/2026

ChatGPT

Home

Day8

Day10

Download file | iLovePDF

github login - Search

gayuaid016-ops/DataSci

localhost:8888/notebooks/Workspace/Day8.ipynb?

Anaconda Toolbox

v4.20.0

Anaconda

Your Toolbox for Python Projects

Anaconda Cloud

Create a New Project

Create a New Notebook

My Projects

Code Snippets

Manage Code Snippets

Environments

Create new Environment

Anaconda AI Assistant

Jupyter Day8 Last Checkpoint: 3 days ago

File Edit View Run Kernel Settings Help

Trusted

JupyterLab Python [conda envbase] *

```
#it means analyzing two variables together to understand the relationship between them
#B1-2
#variate=variables
#Types
#1.variable1- numerical ,numerical (scatterplot)
#2.Variable2-Categorical,numerical (barchart)
#3.Visualization-Categorical,Categorical(Grouped Statistics/Crosstab)

[22]: #Numerical vs Numerical(used to determine whether one numerical variable affects another
plt.figure(figsize=(8,5))
sns.scatterplot(
    x='total_bill',
    y='tip',
    hue='sex',
    data=df
)
plt.title("Total Bill vs Tip")
plt.show()
```

Total Bill vs Tip

sex

Male

Female

10

8

6

bp

Snipping Tool

Screenshot copied to clipboard

Automatically saved to screenshots folder.

Markup and share

28°C Cloudy

Search

12:41 PM 6/9/2026

ChatGPT

Home

Day8

Day10

Download file | iLovePDF

github login - Search

gayuaid016-ops/DataSci

localhost:8888/notebooks/Workspace/Day8.ipynb?

28°C Cloudy

Anaconda

Your Toolbox for Python Projects

Anaconda Cloud

Create a New Project

Create a New Notebook

My Projects

Code Snippets

Manage Code Snippets

Environments

Create new Environment

Anaconda AI Assistant

Jupyter Day8 Last Checkpoint: 3 days ago

File Edit View Run Kernel Settings Help

Code

Python [conda env:base]

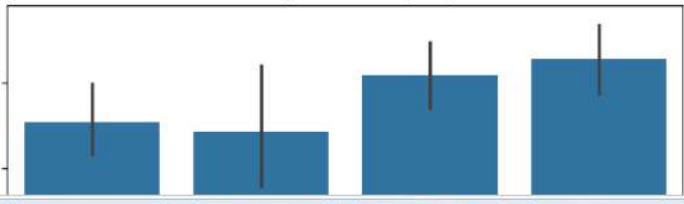
total_bill

10 20 30 40 50

```
[24]: #correlation
corr=df['total_bill'].corr(df['tip'])
print(corr)
0.6757341092113641

[26]: #Categoricals Numerical
#Bar chart
plt.figure(figsize=(8,5))
sns.barplot(
    x='day',
    y='total_bill',
    data=df
)
plt.title("Average Total Bill by Day")
plt.show()
```

Average Total Bill by Day



Snipping Tool

Screenshot copied to clipboard
Automatically saved to screenshots folder.

Markup and share

12:41 PM
6/9/2026

ChatGPT

Home

Day8

Day10

Download file | LLovePDF

github login - Search

gayuaid016-ops/DataSci

localhost:8888/notebooks/Workspace/Day8.ipynb?

Anaconda Toolbox v4.20.0

Anaconda

Your Toolbox for Python Projects

Anaconda Cloud

Create a New Project

Create a New Notebook

My Projects

Code Snippets

Manage Code Snippets

Environments

Create new Environment

Anaconda AI Assistant

Jupyter Day8 Last Checkpoint: 3 days ago

File Edit View Run Kernel Settings Help

Code

JupyterLab Python [conda env:base] * Trusted

Thur Fri Sat Sun

day

[29]:

```
#Grouped Statistics
print(
    df.groupby('day',observed=False)['total_bill'].mean())#->(observed=False for avoiding warning while using statistics)
)
#Multiple statistics
print(
    df.groupby('day',observed=False)['total_bill']
    .agg(['mean','median','min','max'])
)

day
Thur    17.682742
Fri     17.151579
Sat     20.441379
Sun     21.410000
Name: total_bill, dtype: float64
      mean  median   min   max
day
Thur    17.682742   16.20   7.51  43.11
Fri     17.151579   15.38   5.75  40.17
Sat     20.441379   18.24   3.07  50.81
Sun     21.410000   19.63   7.25  48.17
```

[30]:

```
#Categoricalus Categorical(Crosstab -> Libraries)
#Bar chart
pd.crosstab(
    df['sex'],
    df['smoker']
).plot(kind='bar')
plt.title("Sex vs Smoker")
plt.show()
```

Snipping Tool

Screenshot copied to clipboard
Automatically saved to screenshots folder.

Markup and share

29°C Windy

Search

12:41 PM 6/9/2026

ChatGPT

Home

Day8

Day10

Download file | LLovePDF

github login - Search

gayuaid016-ops/DataSci

localhost:8888/notebooks/Workspace/Day8.ipynb?

29°C Windy

12:41 PM 6/9/2026

Anaconda

Your Toolbox for Python Projects

Anaconda Cloud

Create a New Project

Create a New Notebook

My Projects

Code Snippets

Manage Code Snippets

Environments

Create new Environment

Anaconda AI Assistant

Jupyter Day8 Last Checkpoint: 3 days ago

File Edit View Run Kernel Settings Help

JupyterLab Python [conda env:base]

Thun 17.682742 16.20 7.51 43.11
Fri 17.151579 15.38 5.75 40.17
Sat 20.441379 18.24 3.07 50.81
Sun 21.418000 19.63 7.25 48.17

[38]: #Categorical vs Categorical (Crosstab -> Libraries)
#Bar chart
pd.crosstab(
df['sex'],
df['smoker']
) .plot(kind='bar')
plt.title("Sex vs Smoker")
plt.show()

Sex vs Smoker

sex	smoker	count
Male	Yes	60
Male	No	98
Female	Yes	33
Female	No	54

Snipping Tool

Screenshot copied to clipboard
Automatically saved to screenshots folder.

Markup and share

ChatGPT

Home

Day8

Day10

Download file | iLovePDF

github login - Search

gayuaid016-ops/DataSci

localhost:8888/notebooks/Workspace/Day8.ipynb?

29°C Windy

Day8 Last Checkpoint: 3 days ago

File Edit View Run Kernel Settings Help

Code

Python [conda env:base]

Male

Female

sex

```
[31]: #multivariate Analysis
#it means analyzing three or more variables simultaneously to understand complex relationships and patterns in data
#Objectives of multivariate analysis
#1.discover hidden relationships,2.understand interactions among multiple features,3.detect clusters and patterns,
#4.support feature selection,5.prepare data for machine learning
#pairplot-> it compares every numerical feature with every other numerical feature,
#it automatically creates(scatterplot,histogram,relationship matrix)
#why use?-> (Detect correlation,Find cluster,observe distribution,identify

#pairplot
import seaborn as sns
import matplotlib.pyplot as plt
df=sns.load_dataset('iris')
sns.pairplot(
    df,
    hue='species'
)
plt.show()
```

sepal_length

8

7

6

Male

Female

sex

Snipping Tool

Screenshot copied to clipboard

Automatically saved to screenshots folder.

Markup and share

12:41 PM 6/9/2026

ChatGPT

Home

Day8

Day10

Download file | iLovePDF

github login - Search

guyaid016-ops/DataSci

localhost:8888/notebooks/Workspace/Day8.ipynb?

Chat

Anaconda

Your Toolbox for Python Projects

Anaconda Cloud

Create a New Project

Create a New Notebook

My Projects

Code Snippets

Manage Code Snippets

Environments

Create new Environment

Anaconda AI Assistant

Jupyter Day8 Last Checkpoint: 3 days ago

File Edit View Run Kernel Settings Help

Code

petal_wi

sepal_length

sepal_width

petal_length

petal_width

[34]:
#color coded scatter-plot(3d dimension)-> it normally shows xaxis->variable1,yaxis-> variable2
#A third variable can be represented using colors (eg:x-petal length,y-petal width,color-species)-to visualize three numerical variables
plt.figure(figsize=(8,5))
sns.scatterplot(
 x='petal_length',
 y='petal_width',
 hue='species',
 data=df
)
plt.title("Petal length vs petal width")
plt.show()

Petal length vs petal width

species
• setosa
• versicolor
• virginica

idth

Snipping Tool

Screenshot copied to clipboard
Automatically saved to screenshots folder.

Markup and share

29°C
Windy

Search

12:41 PM
6/9/2026

ChatGPT

Home

Day8

Day10

Download file | LLovePDF

github login - Search

gayuaid016-ops/DataSci

localhost:8888/notebooks/Workspace/Day8.ipynb?

Anaconda

Your Toolbox for Python Projects

Anaconda Cloud

Create a New Project

Create a New Notebook

My Projects

Code Snippets

Manage Code Snippets

Environments

Create new Environment

Anaconda AI Assistant

Jupyter Day8 Last Checkpoint: 3 days ago

File Edit View Run Kernel Settings Help

JupyterLab Python [conda env:base]

petal_length

0.0 1 2 3 4 5 6 7

```
[37]: #3D Scatter plot
from mpl_toolkits.mplot3d import Axes3D
import matplotlib.pyplot as plt
fig=plt.figure(figsize=(8,6))
ax=fig.add_subplot(111,projection='3d')
ax.scatter(
    df['sepal_length'],
    df['sepal_width'],
    df['petal_length']
)
ax.set_xlabel('Sepal Length')
ax.set_ylabel('Sepal Width')
ax.set_zlabel('Petal Length')
plt.show()
```

7

6

5

Snipping Tool

Screenshot copied to clipboard
Automatically saved to screenshots folder.

Markup and share

29°C Windy

Search

12:42 PM 6/9/2026

ChatGPT

Home

Day8

Day10

Download file | iLovePDF

github login - Search

gayuaid016-ops/DataSci

localhost:8888/notebooks/Workspace/Day8.ipynb?

Anaconda

Your Toolbox for Python Projects

Anaconda Cloud

Create a New Project

Create a New Notebook

My Projects

Code Snippets

Manage Code Snippets

Environments

Create new Environment

Anaconda AI Assistant

Jupyter Day8 Last Checkpoint: 3 days ago

File Edit View Run Kernel Settings Help

Code

JupyterLab Python [conda env:base]

Sepal Length

6.0 6.5 7.0 7.5 8.0 2.0 2.5

```
[42]: #Correlation-> it measures the strength and direction of relationship between two numerical variables (range -1<=r<=1)
#multicolineraty
import seaborn as sns
import matplotlib.pyplot as plt
df=sns.load_dataset('iris')
print(df)
corr_matrix=df.corr(numeric_only=True)
print(corr_matrix)
plt.figure(figsize=(8,6))
sns.heatmap(
    df.corr(numeric_only=True),
    annot=True,
    cmap='Greens'
)
plt.title("Correlation heatmap")
plt.show()
```

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	setosa
1	4.9	3.0	1.4	0.2	setosa
2	4.7	3.2	1.3	0.2	setosa
3	4.6	3.1	1.5	0.2	setosa
4	5.0	3.6	1.4	0.2	setosa
...	...	...	...	...	...
145	6.7	3.0	5.2	2.3	virginica
146	6.3	2.5	5.0	1.9	virginica
147	6.5	3.0	5.2	2.0	virginica
148	6.2	3.4	5.4	2.3	virginica
149	5.9	3.0	5.1	1.8	virginica

Snipping Tool

Screenshot copied to clipboard
Automatically saved to screenshots folder.

Markup and share

29°C Windy

Search

12:42 PM 6/9/2026

ChatGPT

Home

Day8

Day10

Download file | iLovePDF

github login - Search

gayuaid016-ops/DataSci

localhost:8888/notebooks/Workspace/Day8.ipynb?

29°C Windy

Anaconda

Your Toolbox for Python Projects

Anaconda Cloud

Create a New Project

Create a New Notebook

My Projects

Code Snippets

Manage Code Snippets

Environments

Create new Environment

Anaconda AI Assistant

Jupyter Day8 Last Checkpoint: 3 days ago

File Edit View Run Kernel Settings Help

Code

Python [conda env:base]

petal_width

0.82 -0.37 0.96 1

sepal_length sepal_width petal_length petal_width

[47]:

#VIF(Variance Inflation factor) for multicollinearity detection
#it occurs when independent variables are highly correlated with each other
#it measure how much the variance of a feature is inflated because of its correlation with other features
#VIF Interpretation (1-No correlation,1-5 -Low correlation,5-10 - Moderate multicollinearity,>10-severe Multicollinearity)
from statsmodels.stats.outliers_influence import variance_inflation_factor
X=df.drop("species",axis=1)
vif=pd.DataFrame()
vif["Feature"]=X.columns
vif["VIF"]=[
 variance_inflation_factor(X.values,i)
 for i in range(X.shape[1])
]
print(vif)

	Feature	VIF
0	sepal_length	262.969348
1	sepal_width	96.353292
2	petal_length	172.960962
3	petal_width	55.502860

[50]:

#Remove a High-VIF Feature
#Suppose we remove petal_length
X_new=X.drop("sepal_length",axis=1)
vif_after=pd.DataFrame()
vif_after["Feature"]=X_new.columns

Snipping Tool

Screenshot copied to clipboard
Automatically saved to screenshots folder.

Markup and share

12:42 PM

6/9/2026

ChatGPT

Home

Day8

Day10

Download file | iLovePDF

github login - Search

gayuaid016-ops/DataSci

localhost:8888/notebooks/Workspace/Day8.ipynb?

Anaconda Toolboxv4.20.0

Anaconda

Your Toolbox for Python Projects

Anaconda Cloud

Create a New Project

Create a New Notebook

My Projects

Code Snippets

Manage Code Snippets

Environments

Create new Environment

Anaconda AI Assistant

Jupyter Day8 Last Checkpoint: 3 days ago

File Edit View Run Kernel Settings Help

JupyterLab Python [conda envbase]

```
print(vif)
```

	Feature	VIF
0	sepal_length	262.969348
1	sepal_width	96.353292
2	petal_length	172.960962
3	petal_width	55.502860

[58]:

```
#Remove a High-VIF Feature
#Suppose we remove petal_length
X_new=X.drop("sepal_length",axis=1)
vif_after=pd.DataFrame()
vif_after["Feature"]=X_new.columns
vif_after["VIF"]=[
    variance_inflation_factor(X_new.values,i)
    for i in range(X_new.shape[1])
]
print(vif_after)
```

	Feature	VIF
0	sepal_width	5.856965
1	petal_length	62.871388
2	petal_width	43.292574

[]:

Snipping Tool

Screenshot copied to clipboard
Automatically saved to screenshots folder.

Markup and share

29°C
Windy

Search

12:42 PM
6/9/2026